

Early-Stopping Activation Steering for Factual Preference Alignment in Vietnamese Medical Language Models

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Abstract. Small Language Models (SLMs) offer potential for privacy-preserving clinical decision support but frequently exhibit domain-specific hallucinations in resource-scarce languages like Vietnamese—such as miscalculating pediatric dosages or violating pregnancy contraindications. We propose Early-Stopping Activation Steering, a framework evaluating temporally bounded activation steering on Qwen2.5-7B-Instruct at Layer 8 with Hard Cutoff and Linear Decay schedules restricting vector injection to the first $K=16$ decoding steps ($t \in [1, 16]$, influencing output tokens $y_2 \dots y_{17}$). Under high-cap natural completion (`max_new_tokens=800`), Hard Cutoff ($K=16$) achieves 70.60% reference-preference accuracy (+5.20 pp over baseline, exact McNemar $p = 0.00086$, Holm-adjusted $p_{\text{adj}} = 0.0026$) with 100% EOS hit rate, accompanied by a slight decrease in BERTScore F1 (0.6695 vs 0.6779) and a minor increase in 4-gram repetition (3.74% vs 3.68%). Under bounded-generation stress testing (`max_new_tokens=200`), Linear Decay steering ($K=16$) increases accuracy from 73.00% to 77.20% (+4.20 pp, exact McNemar $p = 0.0055$, Holm-adjusted $p_{\text{adj}} = 0.0165$, 95% Bootstrap CI [+1.40, +7.00] pp), outperforming non-oracle BM25 RAG by +8.60 pp with zero context overhead. Evaluating combined Steering with Hybrid RAG yields 80.40% RefPref accuracy (+1.00 pp observed numerical difference over RAG alone, paired counts $a = 392, b = 5, c = 10, d = 93$, exact McNemar $p = 0.3018$, 95% CI [-0.60, +2.60] pp) while maintaining low 4-gram repetition (3.61%). Descriptive placebo controls ($N=100$ random vectors, $55.82\% \pm 4.15\%$) establish a $+5.15\sigma$ separation above isotropic random direction perturbation.

Keywords: Activation steering · Hallucination mitigation · Clinical language models · Vietnamese NLP · Inference-time intervention · Representation engineering.

1 Introduction

Large Language Models (LLMs) have demonstrated capabilities across healthcare tasks [3]. However, in specialized domains requiring high factual precision—such

as clinical pharmacology and prescription guidance in non-English languages—standard models suffer from domain-specific hallucinations [7]. In Vietnamese clinical settings, these hallucinations range from benign stylistic errors to dangerous medical contradictions, such as recommending contraindicated medications during pregnancy or miscalculating weight-based pediatric dosages.

Traditional mitigation strategies rely on post-hoc supervised fine-tuning (SFT) [10] or Retrieval-Augmented Generation (RAG) [5]. Although RAG provides external knowledge contexts, models may ignore retrieved evidence when internal activation pathways favor non-truthful generation [11], additionally incurring heavy context window overheads and retrieval latencies. Supervised fine-tuning demands substantial computational resources and risks catastrophic forgetting of general abilities [12].

Recently, Representation Engineering (RepE) [1] and Inference-Time Intervention (ITI) [2] have emerged as lightweight, non-destructive alternatives. By extracting a truthfulness steering vector v_{steer} from hidden activation contrasts and injecting it into intermediate layers during generation, activation steering shifts model outputs toward factual accuracy without modifying underlying weights. Related approaches such as Activation Addition (ActAdd) [8] and contrastive activation addition [9] have demonstrated the viability of linear steering for behavioral control.

However, continuous activation steering ($K = \infty$) induces persistent hidden-state norm elevation over long output sequences, associated with cumulative activation displacement. In this paper, we present **Early-Stopping Activation Steering**, a framework evaluating temporally bounded intervention via Hard Cutoff and Linear Decay schedules restricting vector injection to the first $K = 16$ decoding steps ($t \in [1, 16]$), influencing tokens $y_2 \dots y_{17}$.

Our main contributions are: (i) **ViHaluEval-Medical Benchmark**: A benchmark of 14,674 expert-structured Vietnamese clinical Q&A pairs derived from the official Vietnamese National Drug Formulary (2nd Edition, 2018) [6], featuring a question-disjoint test split ($N_{\text{test}} = 500$). (ii) **Dual-Evaluation Paradigm & Pharmacist Dataset Adjudication**: Comprehensive evaluation covering high-cap natural completion (`max_new_tokens=800`, EOS Hit = 99.80%–100.00%) and bounded-generation stress testing (`max_new_tokens=200`), supported by pharmacist dataset adjudication (≥ 5 years experience, Cohen’s $\kappa = 0.91$, Fleiss’ $\kappa = 0.89$) and a 50-sample audit manifest (`human_evaluation_completed_50.csv`) verifying 100% reference accuracy against official drug monographs. (iii) **Empirical Activation-Norm Dynamics**: Hidden-state activation tracking ($\|\hat{h}_8^{(t)}\|_2$) demonstrating persistent norm elevation under continuous steering (56.34 vs 53.39 baseline at $t = 50$, $\Delta = +2.95$) and local Layer 8 residual norm relaxation back to baseline equilibrium (53.39 ± 0.64 , $\Delta = 0.00$) upon hook deactivation. (iv) **Directional Specificity & Multi-Random Placebo Controls**: Negative steering ($-v_{\text{steer}}$, 62.80%), label-permutation placebo steering (v_{shuf} , 68.60%), and $N = 100$ independent random vectors ($55.82\% \pm 4.15\%$ SD) confirming $+v_{\text{steer}}$ (77.20%) achieves an empirical rank of 1/101 ($p_{\text{emp}} = 0.0099$) at $+5.15\sigma$ above random perturbation. (v) **Reproducible Statistical Validation**:

Exact paired 2×2 binomial McNemar testing ($p = 0.0055$, 95% Bootstrap CI [+1.40 pp, +7.00 pp] for bounded stress testing; exact McNemar $p = 0.00086$, 95% Bootstrap CI [+2.40 pp, +8.20 pp] for natural completion). (vi) **Controlled Factorial Ablation:** Controlled 3×3 equal-initial- α_0 matrix alongside a nominal matched cumulative dose control (α_{match}).

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 presents methodology, Section 4 reports experimental results, Section 5 discusses limitations, and Section 6 concludes.

2 Related Work

2.1 Hallucination in Medical Language Models

LLM hallucinations in clinical domains pose unique risks compared to general-purpose applications. Ji et al. [7] survey hallucination phenomena, categorizing them into intrinsic and extrinsic types. In medical settings, Singhal et al. [3] showed that while large models encode clinical knowledge, they remain prone to factual errors in drug interactions, dosage calculations, and contraindications. The HaluEval benchmark [13] introduced evaluation across diverse domains, inspiring our domain-specific ViHaluEval-Medical benchmark for Vietnamese clinical texts. Lin et al. [16] proposed TruthfulQA for measuring model truthfulness, establishing evaluation protocols that informed our contrastive design.

2.2 Inference-Time Intervention and Representation Engineering

Rather than modifying model weights through parameter-efficient fine-tuning [10], inference-time methods intervene directly in internal representations during generation. Li et al. [2] proposed ITI, identifying truthful directions in attention head outputs. Zou et al. [1] generalized this to Representation Engineering (RepE), demonstrating that high-level concepts can be extracted and controlled via linear subspace manipulation. Turner et al. [8] introduced ActAdd, showing that simple vector addition to residual streams steers model behavior. Rimsky et al. [9] extended this with contrastive activation addition. Our work builds upon these foundations but addresses the limitation of continuous steering over long sequences by comparing Hard Cutoff and Linear Decay schedules for temporally bounded steering.

2.3 Evaluation Metrics for Clinical Text Generation

Traditional n-gram overlap metrics like ROUGE [14] provide lexical similarity scores but fail to capture semantic equivalence in multilingual clinical settings. Zhang et al. [15] proposed BERTScore, which computes token-level cosine similarities using contextual BERT embeddings, offering evaluation robust to lexical variation. We adopt a BERTScore-based reference-preference criterion as a proxy for factual preference alignment.

3 Methodology

3.1 ViHaluEval-Medical Benchmark Construction

The ViHaluEval-Medical benchmark comprises 14,674 expert-structured Vietnamese clinical Q&A pairs constructed directly from the official Vietnamese National Drug Formulary (Được thư Quốc gia Việt Nam, 2nd Edition, 2018) [6]. The primary source text spans 14.1 million characters across 1,026 comprehensive drug monographs. To construct contrastive pairs, raw monographs were segmented into 1,000-character context chunks. A vLLM engine powered by **Qwen2.5-7B-Instruct-AWQ** (batch size 64, temperature = 0.7, top- p = 0.9, max tokens = 650) executed a structured prompt. For each chunk, the engine generated distinct clinical question records (q_i), each paired with a pharmacist-verified reference answer (y_i^+) and a hallucinated counter-answer y_i^- (**hallucinated_answer**, drawn from dosage modifications $y_{i,1}^-$, pregnancy safety contradictions $y_{i,2}^-$, or drug interactions $y_{i,3}^-$). Each record in the final manifest (**vietnamese_medical_halueval_15k_specialized.json**) represents an unambiguous triplet (q_i, y_i^+, y_i^-). The dataset spans three clinical categories: Special Dosage Modifications ($N = 5,000$, 34.07%), Contradictory Pregnancy Safety ($N = 4,885$, 33.29%), and Misleading Drug Interactions ($N = 4,789$, 32.64%), strictly summing to $N = 14,674$ total records.

All candidate Q&A pairs underwent manual adjudication by 3 licensed Vietnamese clinical pharmacists (≥ 5 years clinical experience), cross-referencing reference answers against source monographs to verify factual accuracy, achieving high inter-annotator agreement (mean pairwise Cohen’s $\kappa = 0.91$; Fleiss’ $\kappa = 0.89$). Benchmark data is partitioned at the active pharmaceutical ingredient (API) level into a development split ($N_{\text{dev}} = 12,495$ pairs, 85.15%: 70.0% training $N_{\text{train}} = 10,272$ for vector estimation; 15.15% validation $N_{\text{val}} = 2,223$ for hyperparameter tuning) and a held-out test partition ($N_{\text{test_partition}} = 2,179$ pairs, 14.85%), from which $N_{\text{test}} = 500$ evaluation questions were sampled using stratified random sampling with fixed seed 42: Pregnancy Safety ($N = 165$), Drug Interactions ($N = 160$), and Special Dosage ($N = 175$).

3.2 Contrastive Vector Estimation & Directional Controls

Given prompt x_i , reference answer y_i^+ , and counter-answer y_i^- , hidden activations $h_l(x_i, y_i)$ at layer l are collected via forward hook at the final token position of unpadded single-sequence forward passes (batch_size = 1), ensuring [:, -1, :] extracts the true sequence end token without padding interference:

$$h_l(x_i, y_i) = \text{LayerOutput}_l(x_i \oplus y_i)[: , -1, :] \quad (1)$$

The truthfulness steering vector $v_{\text{steer}}^{(l)}$ is computed on $N_{\text{train}} = 10,272$ pairs as normalized mean difference:

$$v_{\text{steer}}^{(l)} = \frac{\mu_l^+ - \mu_l^-}{\|\mu_l^+ - \mu_l^-\|_2} \quad (2)$$

where $\mu_l^\pm = \frac{1}{N} \sum_{i=1}^N h_l(x_i, y_i^\pm)$.

We evaluate four baseline control conditions: (i) **Negative Steering** ($-v_{\text{steer}}$): Steering along anti-truthful direction ($\alpha_0 = -18.0$, RefPref 62.80%). (ii) **Full Label-Permutation** (v_{shuf}): Swapping labels ($y_i^+ \leftrightarrow y_i^-$) across $N_{\text{flipped}} = 5, 198/10, 272$ pairs (RefPref 68.60%). (iii) **Covariance-Matched Distribution** ($N = 20$): Vectors $v_{\text{cov}}^{(r)} = \frac{Lz_r}{\|Lz_r\|_2}$ sampled via Ledoit-Wolf shrinkage matrix factorization $\Sigma_{\text{data}} = LL^T$ from residual difference vectors $d_i = h_i^+ - h_i^-$ on $N_{\text{train}} = 10, 272$ pool (74.81% $\pm$ 0.95%). (iv) **Isotropic Distribution** ($N = 100$): $v_{\text{rand}}^{(r)} = z_r/\|z_r\|_2$ for $z_r \sim \mathcal{N}(0, I_{3584})$ (55.82% $\pm$ 4.15%).

3.3 Early-Stopping Steering Taxonomy & Activation-Norm Dynamics

Residual stream activations $h_l^{(t)}$ at Layer 8 (decoder block index 8 of Qwen2.5-7B) are modified during autoregressive generation ($t \geq 1$) according to:

$$\hat{h}_l^{(t)} = h_l^{(t)} + \alpha(t) \cdot v_{\text{steer}}^{(l)} \quad (3)$$

We evaluate three distinct temporal intervention schedules under fixed α_0 :

$$\alpha_{\text{decay}}(t) = \begin{cases} \alpha_0 \left(1 - \frac{t-1}{K}\right), & 1 \leq t \leq K \\ 0, & t > K \end{cases}, \quad \alpha_{\text{cutoff}}(t) = \begin{cases} \alpha_0, & 1 \leq t \leq K \\ 0, & t > K \end{cases} \quad (4)$$

Continuous Steering ($K = \infty$) applies $\alpha(t) = \alpha_0$ for all $t \geq 1$. Cumulative executed dose depends on temporal window K , initial strength α_0 , and actual generated decoding steps post-prefill $T_{\text{gen}} - 1$. Let $m = \min(K, \max(0, T_{\text{gen}} - 1))$. In the case of early end-of-sequence termination prior to step K , executed dose scales as $D_{\text{cutoff}} = m \cdot |\alpha_0|$, and $D_{\text{decay}} = |\alpha_0| \left[m - \frac{m(m-1)}{2K} \right]$. For full generation ($T_{\text{gen}} \geq 17, K = 16, \alpha_0 = 18.0$), executed dose reaches $D_{\text{cutoff}} = 16 \times 18.0 = 288.0$ under Hard Cutoff, and $D_{\text{decay}} = \frac{17}{2} \times 18.0 = 153.0$ under Linear Decay. Nominal Matched Cumulative Dose Control sets constant per-token coefficient $\alpha_{\text{match}} = 0.0425\alpha_0$ (0.7650 for $\alpha_0 = 18.0$), scaling executed dose with generated sequence length:

$$D_{\text{executed}} = (T_{\text{gen}} - 1) \cdot |\alpha_{\text{match}}| \quad (5)$$

3.4 BERTScore Reference-Preference Criterion & Repetition Metric

Factual preference alignment is evaluated via BERTScore [15] comparisons against reference answer y^+ and counter-answer y^- :

$$\text{RefPref}(\hat{y}) = \mathbb{1}[\text{BS}(\hat{y}, y^+) > \text{BS}(\hat{y}, y^-)] \quad (6)$$

where $\text{BS}(\cdot, \cdot)$ denotes BERTScore F1 using **bert-base-multilingual-cased** (version 0.3.13, layer 9, default rescaling/IDF disabled). Text repetition is measured via word-level 4-gram repetition (**Rep-4**):

$$\text{Rep-4}(\hat{y}_i) = \begin{cases} 100 \times \left(1 - \frac{|\text{unique 4-grams in } G_4(\hat{y}_i)|}{|\text{total 4-grams in } G_4(\hat{y}_i)|}\right), & |G_4(\hat{y}_i)| > 0 \\ 0.0\%, & |G_4(\hat{y}_i)| = 0 \end{cases} \quad (7)$$

where $G_4(\hat{y}_i)$ denotes consecutive word 4-grams extracted via whitespace tokenization (`text.split()`), aggregated as sample macro-average across $N = 500$ sequences. Across all 4,000 generated test outputs (2,000 natural completion + 2,000 bounded stress test), zero outputs (0.00%) exceeded the 512-subword evaluator window limit (0% truncation rate) and zero score ties (0.00%) occurred.

3.5 Experimental Setup & Implementation Protocol

All experiments evaluate Qwen2.5-7B-Instruct [4] loaded in 4-bit NormalFloat4 (NF4) double-quantization via BitsAndBytes 0.42.0 on NVIDIA Tesla T4 GPUs (CUDA 12.1, PyTorch 2.2.0, Transformers 4.38.2). All generation runs use deterministic greedy decoding (`temperature=0.0`, `do_sample=False`, `top_p=1.0`, seed 42) with key-value caching enabled (`use_cache=True`). ChatML template system instruction: “*Bạn là một chuyên gia y được lâm sàng Việt Nam. Hãy trả lời câu hỏi y khoa dưới đây một cách chính xác, khách quan dựa trên Dược thư Quốc gia Việt Nam (2nd Edition, 2018).*”. PyTorch forward hook attached to decoder block `model.model.layers[8]` handles prefill ($t = 0$, sequence length > 1 , resetting `self.t=0`) and single-token step decoding ($t \geq 1$, sequence length $= 1$, incrementing `self.t+=1`):

```
class SteeringHookHandler:
    def __init__(self, v_steer, alpha_0=18.0, K=16, schedule="linear_decay"):
        self.v_steer, self.alpha_0, self.K, self.schedule, self.t = v_steer, alpha_0, K,
            schedule, 0
    def reset(self): self.t = 0
    def hook_fn(self, module, input, output):
        out = output[0] if isinstance(output, tuple) else output
        if out.shape[1] > 1: self.t = 0; return output
        self.t += 1
        a_t = self.alpha_0 * max(0.0, 1.0 - (self.t-1)/self.K) if self.schedule == "
            linear_decay" else (self.alpha_0 if self.t <= self.K else 0.0) if self.schedule
            == "hard_cutoff" else (self.alpha_0 if self.schedule == "continuous" else 0.0)
        if a_t != 0.0: out[:, -1, :] += a_t * self.v_steer.to(device=out.device, dtype=out.
            dtype)
        return output
```

Retrieval-Augmented Generation (RAG) uses 14,576 passages (180 ± 45 words / 215 ± 52 Qwen subword tokens) from the drug formulary [6], evaluating Lexical BM25 (`pyvi.ViTokenizer`, $k_1 = 1.5$, $b = 0.75$), Dense Retrieval (BAAI/bge-m3, 1024-d cosine similarity), and Hybrid RRF ($k_{\text{rrf}} = 60$). Top-1 passage ($k = 1$) is formatted into ChatML context prompt. Hyperparameter selection (layer $l = 8$, cutoff $K = 16$, scale $\alpha_0 = 18.0$) was performed on validation split ($N_{\text{val}} = 2,223$, $N_{\text{eval}} = 500$). Complete raw output manifests (`evaluation_logs.jsonl`), per-sample generated texts, test split indices (`test_ids.json`), steering vector weights (`v_steer.pt`), and evaluation scripts are available in the public project repository (<https://github.com/anonymous-submission/early-stopping-steering>).

4 Results

4.1 Primary Natural Completion Evaluation ($\text{max_new_tokens}=800$)

Under high-cap natural completion ($\text{max_new_tokens}=800$), models achieve EOS Hit rates of 99.80%–100.00%. As shown in Table 1, Hard Cutoff ($K=16$) achieves peak **70.60% RefPref accuracy** (+5.20 pp over baseline, Holm-adjusted $p_{\text{adj}} = 0.0026$) with 100% EOS hit rate, accompanied by a slight decrease in BERTScore F1 (0.6695 vs 0.6779) and a minor increase in 4-gram repetition (3.74% vs 3.68%). In each table, bold values highlight the highest score per metric column.

Table 1. Primary natural completion evaluation ($N_{\text{test}} = 500$, $\text{max_new_tokens}=800$) and paired statistical significance vs unsteered baseline (65.40%).

Condition	RefPref (%)	BERT F1	EOS (%)	Rep-4 (%)	McNemar p	95% CI (pp)
Unsteered Baseline	65.40	0.6779	100.00	3.68	—	—
Continuous ($K=\infty$)	68.60	0.6705	99.80	3.50	0.0365	[+0.40, +6.00]
Linear Decay ($K=16$)	67.80	0.6732	99.80	3.74	0.1189	[−0.20, +5.20]
Hard Cutoff ($K=16$)	70.60	0.6695	100.00	3.74	0.00086	[+2.40, +8.20]

Note: Paired contingency counts (a, b, c, d): Hard Cutoff (311, 16, 42, 131), Continuous (309, 18, 34, 139), Linear Decay (308, 19, 31, 142), where b : baseline win, c : steered win. 95% CIs via $B = 10,000$ bootstrap. Rep-4 evaluated via word-level 4-gram whitespace macro-average.

4.2 Controlled Bounded-Generation Stress Test ($\text{max_new_tokens}=200$)

Under bounded stress testing ($\text{max_new_tokens}=200$), Linear Decay steering ($K=16$) achieves the highest accuracy among tested configurations at **77.20%** (386/500, +4.20 pp over baseline 73.00%, exact McNemar $p = 0.0055$, Holm $p_{\text{adj}} = 0.0165$, 95% CI [+1.40, +7.00] pp), representing a net difference of 1 question over Hard Cutoff (77.00%, 385/500), while 4-gram repetition increases slightly to 3.74% (vs 3.67% baseline, Table 2).

4.3 Empirical Activation-Norm Dynamics & Teacher-Forcing Measurement

To quantify residual stream magnitude changes without autoregressive token divergence, we measure activation norm drift under single-step teacher forcing across $N = 500$ prompts:

$$\Delta_{\text{norm}}(t) = \frac{1}{N} \sum_{i=1}^N \left(\|\hat{h}_8^{(t)}(x_i)\|_2 - \|h_8^{(t)}(x_i)\|_2 \right) \quad (8)$$

Table 2. Primary bounded stress test evaluation ($N_{\text{test}} = 500$, `max_new_tokens`=200, Layer 8) and paired statistical significance vs unsteered baseline (73.00%).

Condition	RefPref (%)	BERT F1	ROUGE-L (%)	Rep-4 (%)	McNemar p	95% CI (pp)
Unsteered Baseline	73.00	0.7133	23.12	3.67	–	–
Continuous ($K=\infty$)	75.60	0.7133	23.36	3.90	0.0854	$[-0.20, +5.40]$
Hard Cutoff ($K=16$)	77.00	0.7151	23.16	3.89	0.0078	$[+1.20, +6.80]$
Linear Decay ($K=16$)	77.20	0.7150	23.21	3.74	0.0055	$[+1.40, +7.00]$

Note: Paired contingency counts (a, b, c, d): Linear Decay (349, 16, 37, 98), Hard Cutoff (349, 16, 36, 99), Continuous (347, 18, 31, 104). 95% CIs via $B = 10,000$ bootstrap. Rep-4 evaluated via word-level 4-gram whitespace macro-average.

Unsteered baseline generation establishes reference norm equilibrium $\|h_g^{(t)}\|_2 = 53.39 \pm 0.64$ ($t = 50$). Continuous steering ($K = \infty$) induces persistent norm elevation ($\Delta_{\text{norm}}^{(50)} = +2.95$). Extending injection past $K = 16$ accumulates active drift during intervention ($K = 24$: $\Delta_{\text{norm}}^{(20)} = +1.12$; $K = 32$: $\Delta_{\text{norm}}^{(20)} = +1.98$). Deactivating injection past step K directly restores Layer 8 residual outputs to baseline magnitude (53.39 ± 0.64 , $\Delta_{\text{norm}}^{(50)} = 0.00$), preventing persistent magnitude displacement locally at Layer 8 without active state recovery.

4.4 Directional Specificity Controls, Placebo Baselines & RAG Comparisons

Table 3 reports performance across controls and multi-retriever RAG. Negative steering ($-v_{\text{steer}}$) reduces RefPref to 62.80%. Isotropic random controls ($N = 100$, $55.82\% \pm 4.15\%$) confirm $+v_{\text{steer}}$ (77.20%) achieves an empirical rank of 1/101 ($p_{\text{emp}} = 0.0099$) at $+5.15\sigma$ separation. Linear Decay steering (+8.60 pp over BM25 RAG 68.60%, exact McNemar $p = 6.11 \times 10^{-7}$, paired counts $b = 16, c = 59$) achieves accuracy competitive with Hybrid RRF RAG (76.20%, McNemar $p = 0.5682$, paired counts $b = 22, c = 27$) without context window expansion.

4.5 Controlled Factorial Ablation Matrix

A 3×3 factorial ablation (Table 4) holding initial α_0 fixed across schedules alongside Nominal Matched Dose ($\alpha_{\text{match}} = 0.0425\alpha_0$) confirms Linear Decay ($K = 16$) achieves highest accuracy at $\alpha_0 = 18.0$ (77.20% RefPref, 3.74% Rep-4).

4.6 Temporal Shift, Prefix Disentanglement & Combined RAG Evaluation

Temporal Window Injection Shift Control. Intervening early ($t \in [1, 16]$) under Hard Cutoff step injection ($\alpha_0 = 18.0$) achieves 3.89% Rep-4 at 200 tokens (3.74% at 800 tokens), whereas applying the Linear Decay schedule

Table 3. Comparison against directional controls, placebo baselines, and multi-retriever RAG ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$).

Method / Condition	Raw Count	RefPref (%)	Recall@1	MRR	BERT F1	Latency (s) [†]
Isotropic Gaussian ($N=100$)	–	55.82 ± 4.15	–	–	0.6945	23.33 ± 0.30
Negative Steered ($-v_{\text{steer}}$)	314 / 500	62.80	–	–	0.6889	23.34 ± 0.28
BM25 Lexical RAG	343 / 500	68.60	19.20%	0.2433	0.6970	7.32 ± 0.45
Label-Shuffled (v_{shuf})	343 / 500	68.60	–	–	0.6845	23.34 ± 0.28
Unsteered Baseline	365 / 500	73.00	–	–	0.7133	23.33 ± 0.30
Cov-Matched ($N=20$)	–	74.81 ± 0.95	–	–	0.7182	23.34 ± 0.28
Dense BGE-M3 RAG	374 / 500	74.80	42.60%	0.5124	0.7180	7.27 ± 0.42
Hybrid RRF RAG	381 / 500	76.20	48.20%	0.5681	0.7190	7.39 ± 0.48
Linear Decay Steered	386 / 500	77.20	–	–	0.7150	23.34 ± 0.28
Oracle Gold-Context RAG	447 / 500	89.40	100.0%	1.0000	0.8002	7.17 ± 0.38

Note: Paired McNemar counts vs Steering ($+v_{\text{steer}}$): vs BM25 RAG ($b = 16$ BM25 win, $c = 59$ Steering win, $p = 6.11 \times 10^{-7}$); vs Hybrid RAG ($b = 22$ Hybrid win, $c = 27$ Steering win, $p = 0.5682$). [†] Profiled in separate Kaggle T4 sessions.

Table 4. Controlled factorial ablation matrix under matched initial steering strength ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$).

α_0	Schedule	RefPref (%)	BERTScore F1	ROUGE-L (%)	Rep-4 (%)
–	Unsteered Baseline	73.00	0.7133	23.12	3.67
15.0	Continuous ($K=\infty$)	75.20	0.7136	23.43	3.65
	Hard Cutoff ($K=16$)	75.80	0.7146	23.11	3.73
	Linear Decay ($K=16$)	75.60	0.7142	23.10	4.03
	Nominal Matched Dose	75.40	0.7144	23.40	3.78
18.0	Continuous ($K=\infty$)	75.60	0.7133	23.36	3.90
	Hard Cutoff ($K=16$)	77.00	0.7151	23.16	3.89
	Linear Decay ($K=16$)	77.20	0.7150	23.21	3.74
	Nominal Matched Dose	75.40	0.7140	23.21	4.38
20.0	Continuous ($K=\infty$)	75.40	0.7134	23.37	4.05
	Hard Cutoff ($K=16$)	76.40	0.7139	23.17	3.87
	Linear Decay ($K=16$)	76.60	0.7147	23.15	3.74
	Nominal Matched Dose	75.20	0.7136	23.33	4.52

($K = 16$) yields 3.74% at 200 tokens (3.74% at 800 tokens). Delayed step injection ($t \in [17, 32]$) yields 3.92% Rep-4 at 800 tokens, and later injection ($t \in [33, 48]$) yields 72.80% RefPref, which is indistinguishable from the unsteered baseline (73.00%), representing a net difference of 1 question out of 500 items ($p = 1.000$).

Prefix Length Disentanglement Analysis. Truncated prefix evaluation (Table 5) confirms persistent positive RefPref gains (+4.00 to +5.40 pp over baseline) across sequence boundaries ($T \in \{50, 100, 200, 400, 800\}$ tokens), demonstrating persistent relative ranking over baseline across truncation boundaries.

Table 5. Truncated prefix RefPref accuracy (%) across sequence length boundaries ($N_{\text{test}} = 500$).

Schedule / Condition	$T=50$	$T=100$	$T=200$	$T=400$	$T=800$
Unsteered Baseline	71.20	72.00	73.00	67.80	65.40
Continuous ($K=\infty$)	73.80	74.20	75.60	71.40	68.60
Hard Cutoff ($K=16$)	75.80	76.40	77.00	73.20	70.60
Net Gain over Base (pp)	+4.60	+4.40	+4.00	+5.40	+5.20

Pharmacist Dataset Adjudication & Metric Proxy Scope. Benchmark reference answers (y^+) were constructed directly from official drug monographs [6] and adjudicated by 3 licensed clinical pharmacists (≥ 5 years experience), achieving high inter-annotator agreement ($\kappa = 0.91$, Fleiss’ $\kappa = 0.89$). A stratified random audit sample of $N = 50$ test questions (`human_evaluation_completed_50.csv`) confirmed 100% factual correctness of reference answers against source monographs (mean correctness score 2.0/2.0). Automated BERTScore RefPref evaluates model outputs relative to these expert-verified ground-truth answers. We explicitly treat RefPref as an automated semantic preference proxy, acknowledging that direct clinician-adjudicated scoring of generated model completions across clinical evaluations remains essential future work.

Combined Hybrid RAG + Steering Evaluation. Under natural completion (`max_new_tokens=800`), evaluating combined Hybrid RAG + Steering (Linear Decay $K = 16, \alpha_0 = 18.0$) across full $N_{\text{test}} = 500$ samples achieves **80.40% RefPref accuracy** (+1.00 pp observed numerical difference over standalone Hybrid RAG 79.40%, paired counts $a = 392, b = 5, c = 10, d = 93$, exact binomial McNemar $p = 0.3018$, 95% Bootstrap CI $[-0.60 \text{ pp}, +2.60 \text{ pp}]$). Combined RAG + Steering maintains low 4-gram repetition Rep-4 (**3.61%** vs 3.78% RAG alone, 3.68% Baseline) while producing concise clinical completions (167.86 tokens vs 171.82 tokens RAG alone).

5 Discussion & Limitations

Reference-preference accuracy is a semantic proxy for factual alignment rather than a clinician-adjudicated correctness rate. Equal initial α_0 produces unequal cumulative doses across schedules (Section 3.3). Validation tuning ($N_{\text{eval}} = 500$) identified highest performance at Layer 8 (77.40%), cutoff $K = 16$ (77.40%), and scale $\alpha_0 = 18.0$ (77.40%). Deactivating vector injection past step K directly restores Layer 8 outputs to unsteered baseline magnitude (53.39 ± 0.64 , $\Delta_{\text{norm}}^{(50)} = 0.00$), preventing persistent norm elevation ($\Delta = +2.95$ under continuous steering) locally at Layer 8, while full network residual propagation remains a structural hypothesis. Percentile bootstrap CIs ($B = 10,000$) use paired question-index resampling. Isotropic placebo controls ($N = 100$, $55.82\% \pm 4.15\%$) confirm $+v_{\text{steer}}$ (77.20%) achieves an empirical rank of 1/101 ($p_{\text{emp}} = 0.0099$) at $+5.15\sigma$ separation. Under high-cap natural completion (`max_new_tokens=800`), Hard Cutoff ($K = 16$) maintains 100.00% EOS hit rate while improving preference accuracy (+5.20 pp over baseline: 70.60% vs 65.40%, exact binomial McNemar $p = 0.00086$, $p_{\text{adj}} = 0.0026$, 95% Bootstrap CI [+2.40 pp, +8.20 pp]), accompanied by a slight decrease in BERTScore F1 (0.6695 vs 0.6779) and a minor increase in 4-gram repetition (+0.06 pp, 3.74% vs 3.68%). Continuous steering at 800 tokens exhibits 3.50% Rep-4, indicating early stopping does not reduce repetition compared to continuous steering at long output lengths, but rather avoids continuous norm elevation while achieving higher preference accuracy.

6 Conclusion

We introduced Early-Stopping Activation Steering, a temporally bounded inference-time intervention for Vietnamese clinical Small Language Models. Evaluated on ViHaluEval-Medical (14,674 records derived from the Vietnamese National Drug Formulary, 2nd Edition, 2018), restricting steering to initial $K = 16$ decoding steps yields reference-preference alignment with zero model-weight updates and zero retrieval latency. Under natural completion (`max_new_tokens=800`), Hard Cutoff ($K = 16$) achieves 70.60% RefPref accuracy (+5.20 pp, exact McNemar $p = 0.00086$, $p_{\text{adj}} = 0.0026$). Under bounded stress testing (`max_new_tokens=200`), Linear Decay ($K = 16$) achieves 77.20% (+4.20 pp, $p = 0.0055$). Compared to non-oracle BM25 RAG (68.60%), early-stopping steering provides +8.60 pp superior accuracy ($p = 6.11 \times 10^{-7}$), offering a practical alignment mechanism warranting clinician-adjudicated validation.

Acknowledgments. The author acknowledges FPT University Quy Nhon for facilitating computational resources and academic support for this research project.

Disclosure of Interests. The author declares no competing financial or non-financial interests directly related to the work described in this paper.

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